

Formal Definition: Function of Two Variables

Definition (Function of Two Variables)

Let $z = f(x, y)$ be a function with two variables x and y .

- x and y are the **independent variables**.
- z is the **dependent variable**.
- Unless otherwise specified, the **domain** D is the **largest subset of $\mathbb{R}^2$** on which f is mathematically well-defined.

Boundary Convention for Domains in $\mathbb{R}^2$

- **Strict inequality ($>$ or $<$):** Boundary drawn as a **dashed curve** (points excluded).
- **Non-strict inequality ($\geq$ or $\leq$):** Boundary drawn as a **solid curve** (points included).



Dashed boundary vs. Solid boundary